

the outer radius of the low-velocity zone, substituted for r_p , to find the distance and time the ray traveled on its way down to the low-velocity zone. Subtracting these from the known $T(\Delta)$ curve gives a $T'(\Delta)$ curve for a “mini-earth” with radius r' . Because in this “mini-earth” the velocity increases with depth, the Herglotz–Wiechert method is applied again.

3.5 Body wave travel time studies

We saw in the last section that travel time data can be used to determine seismic velocity as a function of depth. Beginning early in the 1900s, travel time tables were compiled by combining data from many earthquakes observed at various epicentral distances. These seismological observations provide the primary data for our view of the basic features of the earth’s velocity structure. This picture, an essentially layered earth composed of a thin crust, a mantle, a liquid outer core, and a solid inner core, is key to our thinking about how the earth evolved and operates. This concept was largely developed by the 1940s, as illustrated in Fig. 3.5-1, showing the classic Jeffreys–Bullen¹ (JB) earth model. The JB model treated the earth as a series of shells, characterized by the behavior of the velocity with depth (Table 5.1-1). The mantle was divided into an *upper mantle* (region B) and a *lower mantle* (region D),

Table 3.5-1 Regions in Jeffreys–Bullen earth model.

Region	Depth (km)	Features of region
A	33	Crustal layers
B	413	Upper mantle: steady positive P and S velocity gradients
C	984	Mantle transition region
D	2898	Lower mantle: steady positive P and S velocity gradients
E	4982	Outer core: steady positive P velocity gradient
F	5121	Core transition: negative P velocity gradient
G	6371	Inner core: small positive P velocity gradient

Source: After Bullen and Bolt (1985).

both of which had smooth velocity gradients. These upper and lower mantle regions were separated by region C, the *mantle transition zone* where velocities increase rapidly with depth. Below the *core–mantle boundary* (CMB), the core was divided into an *outer core* (region E) and an *inner core* (region G), separated by a transition zone (region F). The *inner core boundary* (ICB) separated regions F and G. Subsequently, the lower mantle was divided into regions D’ (1000–2700 km depth), most of the lower mantle with a smooth velocity gradient, and D’’ (2700–2900 km), the zone above the core–mantle boundary with a reduced velocity gradient.

Subsequent studies have derived models, such as the IASP91 model, also shown in Fig. 3.5-1, which confirm the basic structure of the JB model and provide better resolution of important regions. For example, the JB model did not resolve shear velocities in the inner core, whereas recent models have finite S velocity in the inner core, implying that it is solid. Similarly, recent models provide more details about the mantle transition zone and the core–mantle boundary, and do not include the velocity “notch” at the inner core–outer core boundary.

Jeffreys’ and Bullen’s derivation of a radially symmetric earth model from travel time observations converted the previous crude picture of the earth into one that has since changed only in detail. More recent radial velocity models do not differ much from each other, so they are likely to be converging on an accurate radial model for the earth. Such average, or *reference*, models and travel time curves, such as JB, IASP91, and PREM (for Preliminary Reference Earth Model, Section 3.8), are derived from data around the world and so average over local variations in structure. Regional differences can then be viewed as perturbations relative to a reference model.

However, lateral differences in structure can be significant and provide insight into tectonic processes. Thus a major current goal of seismology is to define the three-dimensional velocity structure that results from the fact that the earth is a

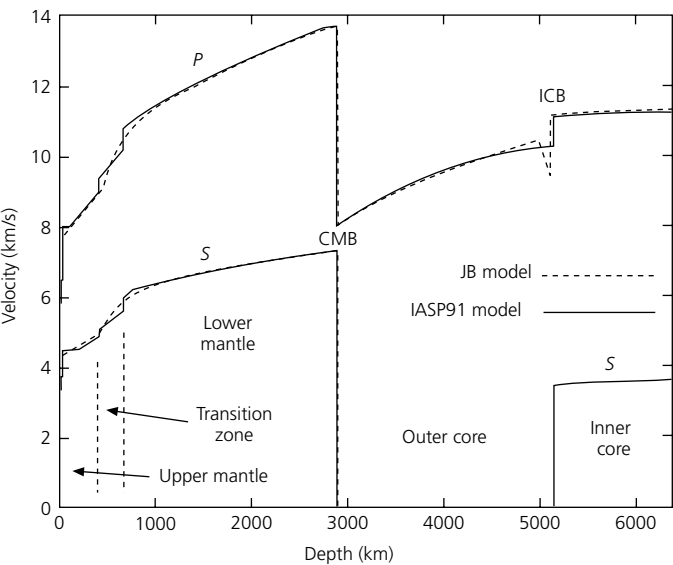


Fig. 3.5-1 Comparison of the classic Jeffreys–Bullen earth model (Jeffreys and Bullen, 1940) and a newer model, IASP91 (Kennett and Engdahl, 1991). Although IASP91 and its successor, AK135 (Kennett *et al.*, 1995), have improved resolution in the mantle transition zone and the core, the newer models are generally similar to that derived using hand-cranked calculators.

¹ The model was derived from extensive joint research into earth structure by Sir Harold Jeffreys (1891–1989), who established in 1926 that the core was liquid, and Keith Bullen (1906–76).

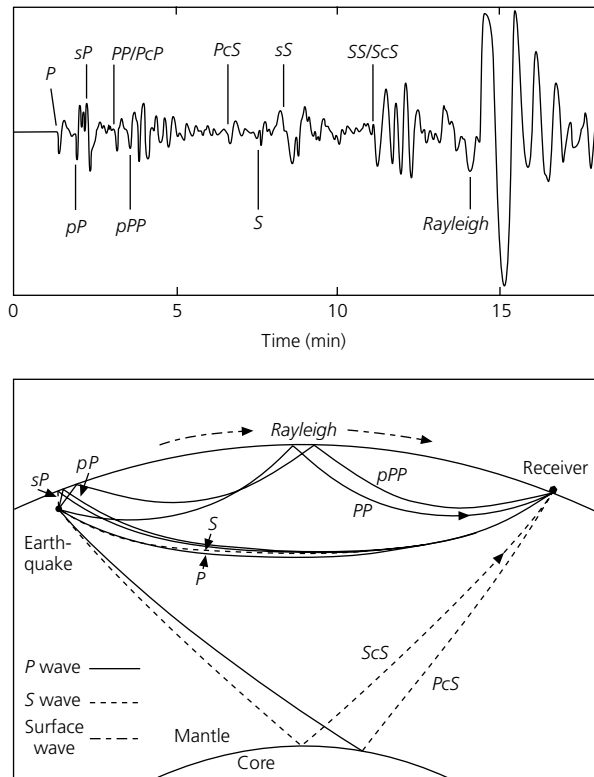


Fig. 3.5-2 *Top*: Long-period vertical component seismogram at Golden, Colorado, showing various seismic phases. *Bottom*: Ray paths for some of the seismic phases labeled on the seismogram. Paths taken as *P* waves are shown as solid lines; paths taken as *S* waves are shown as dashed lines. Although *P* and *S* are both direct phases, they do not travel the exact same path because their velocities differ. Similarly, the ray path for *PcS* is asymmetric, and *pP* and *sP* do not reflect off the surface at the same location.

geologically active planet. Convection in the earth causes three-dimensional temperature variations that result in observable velocity variations. In addition, mantle flow appears to generate seismic anisotropy at the top and the bottom of the mantle, and magnetic stresses due to outer core flow may cause inner core anisotropy. Resolving this three-dimensional structure requires sophisticated analysis techniques. For example, travel time studies are complemented by waveform modeling, and stacking techniques are applied to enhance seismic signals. The suggested reading provides some reviews of recent studies.

This section focuses on determining velocity structure, so we largely defer discussion of the chemical, mineralogical, thermal, and rheological factors that cause these variations for later sections.

3.5.1 Body wave phases

We have seen that seismic waves can travel between a source and a receiver along multiple paths. For example, increases in velocity can cause triplications, yielding three distinct arrivals

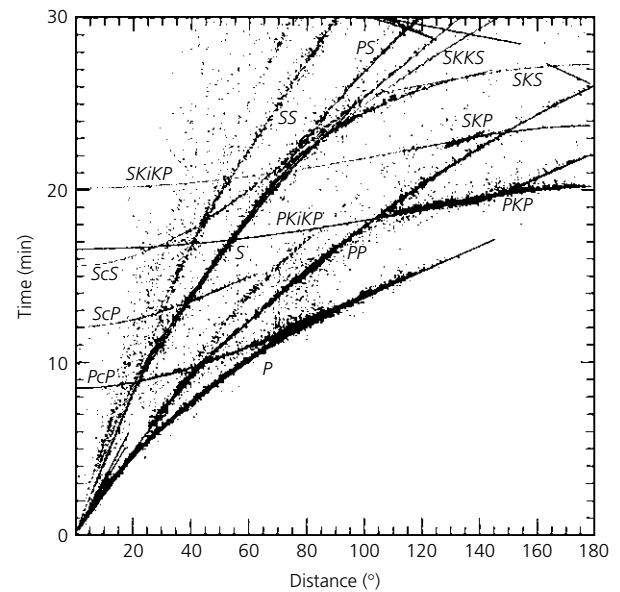


Fig. 3.5-3 Travel time data for various body wave phases and travel time curves for model IASP91. The travel times are corrected to those for an earthquake at the surface. The data are 57,655 travel times from 104 sources (earthquakes and explosions). (Kennett and Engdahl, 1991.)

at a receiver. Multiple reflections off various layers and diffractions can bring additional arrivals. Hence seismograms contain many arrivals, or *phases*, corresponding to different travel paths. This is illustrated by Fig. 3.5-2, discussed in Section 1.1, showing a few of the phases that are observed and some of the corresponding ray paths. All the phases shown, except for the Rayleigh surface wave, are *body waves* that travel through the earth's interior.

Such seismograms provide the observations that are combined to generate travel time tables. Figure 3.5-3 illustrates the process; the dots are travel times observed at various epicentral distances for a set of earthquakes and nuclear explosions. The data define lines giving the travel times of different phases. Such observations can be used to develop and test earth models giving *P* and *S* velocities as a function of depth. These models predict the observed travel times quite well, as shown by the fit of the theoretical travel times (lines in Fig. 3.5-3) to the observations. The travel times depend on the source depth, as shown in Fig. 3.5-4 for a surface source and a source at 600 km depth.

Although the details of an earth model depend on the specific data used to construct it, the key features of IASP91 are characteristic of recent models. The model represents a global average of the velocity structure that varies somewhat between locations. The crust is 35 km thick, an average between thin oceanic and thick continental crust (Fig. 3.2-17). Velocities increase smoothly through the upper mantle, to a depth of 410 km. The mantle transition zone, from about 400–700 km depth, contains depth intervals near 410 km and 660 km

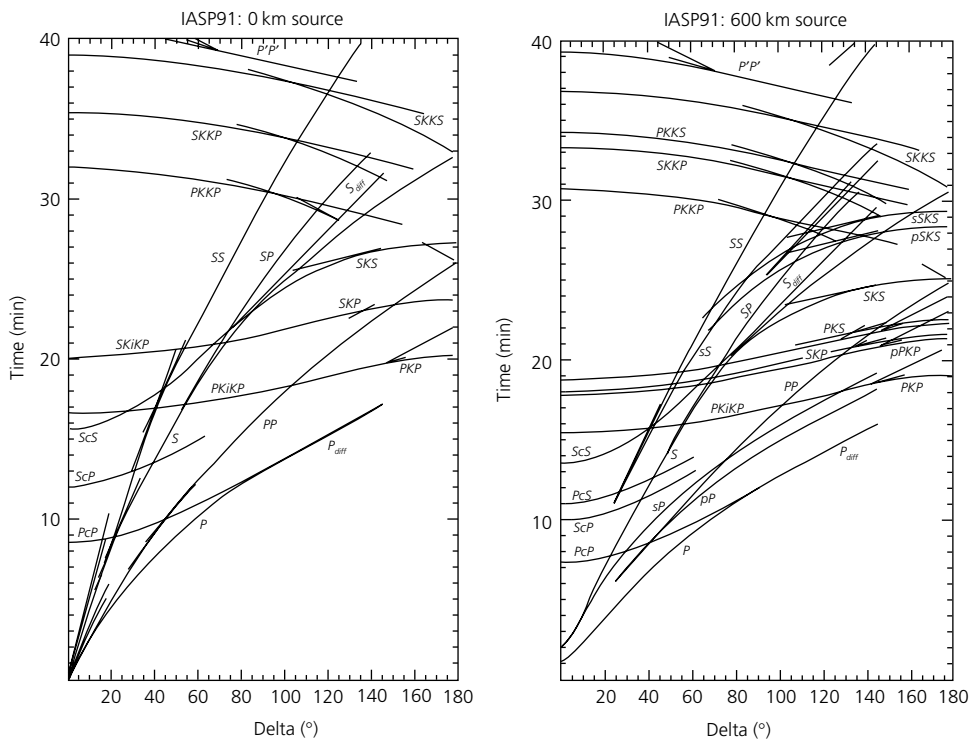


Fig. 3.5-4 Travel time curves for body wave phases for model IASP91 computed for an earthquake at the surface (*left*) and at a focal depth of 600 km (*right*). (Kennett and Engdahl, 1991.)

where the velocities increase rapidly.² Although these regions are often referred to as the 410 km and 660 km discontinuities, their exact depths vary from place to place. From about 700 to 2890 km depth the velocities increase smoothly throughout the lower mantle. At about 2890 km, the *P* velocity drops sharply, and the *S* velocity goes to zero, corresponding to the liquid outer core. The outer core extends to a depth of about 5150 km, beneath which the solid inner core has higher velocities, including a finite *S*-wave velocity. As we will see, these variations in velocity with depth are thought to reflect important changes in the physical, chemical, thermal, and mineralogical state of the materials present.

Seismic phases are named, based on their paths through the earth (Fig. 3.5-5, Table 3.5-2). The direct *P*-wave and *S*-wave arrivals are denoted “*P*” and “*S*.” Another class of arrivals involve reflections at the earth’s surface. The *P*-wave arrival corresponding to a single surface reflection is called *PP*, that for two reflections is *PPP*, and so on. Similarly, *SS* and *SSS* correspond to *S* waves reflected at the surface. Because *P* waves can convert to *S* waves, and vice versa, *PS* is a *P* wave converted to an *S* wave upon surface reflection, and *SP* is the reverse. Consideration of the ray paths shows that the travel time for *PP* at a given distance should be twice the travel time of *P* at half that distance — that is, to a point midway between the source and

Table 3.5-2 Body wave phase nomenclature.

Name	Description
<i>P</i>	Compressional wave
<i>S</i>	Shear wave
<i>K</i>	<i>P</i> wave through outer core
<i>I</i>	<i>P</i> wave through inner core
<i>J</i>	<i>S</i> wave through inner core
<i>PP</i>	<i>P</i> wave reflected at surface
<i>PPP</i>	<i>P</i> wave reflected at surface twice
<i>SP</i>	<i>S</i> wave reflected at surface as <i>P</i> wave
<i>PS</i>	<i>P</i> wave reflected at surface as <i>S</i> wave
<i>pP</i>	<i>P</i> wave upgoing from focus, reflected at surface
<i>sP</i>	<i>S</i> wave upgoing from focus, converted to <i>P</i> at surface
<i>c</i>	Wave reflected at core–mantle boundary (e.g., <i>ScS</i>)
<i>i</i>	Wave reflected at inner core–outer core boundary (e.g., <i>PKiKP</i>)
<i>p’</i>	Abbreviation for <i>PKP</i>
<i>P_d</i> or <i>P_{diff}</i>	<i>P</i> wave diffracted along core–mantle boundary

Source: After Bolt (1982).

the receiver. Similarly, the travel time for *PPP* should be three times the travel time for one-third the distance.

The surface-reflected phases *PP* and *SS* (as well as *SSS*, *SSSS* or *S4*, etc.) have unusual characteristics. By Fermat’s principle (Section 2.5.9), seismic phases have either minimum or maximum travel times with respect to adjacent paths. Most arrivals (*P*, *S*, *pP*, *ScS*, etc.) are minimum-time phases, but the surface

² Although early estimates put the locations of the discontinuities at depths of 400 and 670 km, recent revisions place them closer to 410 and 660 km. We will use the differing values interchangeably.

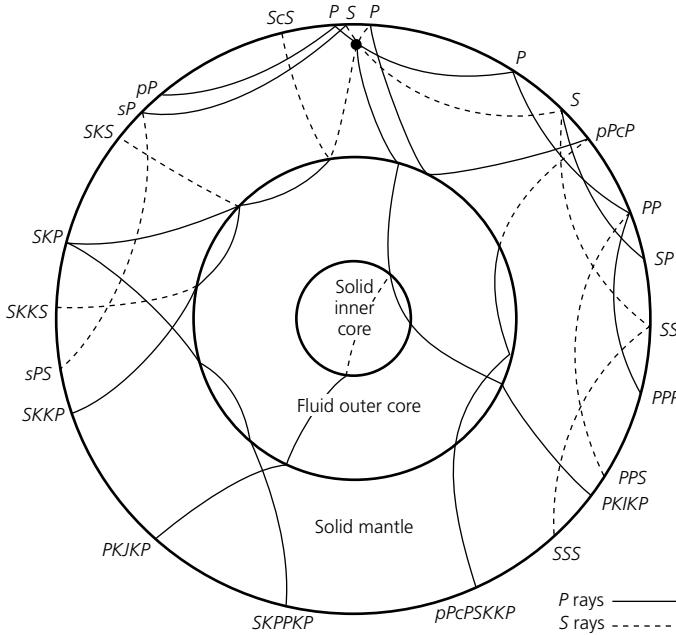


Fig. 3.5-5 Examples of body wave phases illustrating the nomenclature used. “P” and “S” designate direct ray paths, whereas “p” and “s” denote upgoing paths from the earthquake. Hence SP designates an S wave through the mantle reflected at the surface as P. “c” designates a reflection at the core–mantle boundary, so PcP is a P wave reflected at the core, and PcS is a P wave reflected as S. “K” and “I” denote P waves that traveled through the outer and inner cores, and “i” designates a reflection at the inner core’s boundary. Hence PKIKP travels through the mantle, outer core, and inner core. PKJKP, which travels as S through the inner core, has only recently been conclusively observed. (After Bolt, 1982. From *Inside the Earth* by Bruce A. Bolt. © 1982 by W. H. Freeman and Company. Used with permission.)

reflections are maximum-time phases with respect to distance. To see this, consider ray paths for a surface reflection that differ slightly from the true path, so the reflection bounces off the surface a small distance ϵ from the actual bounce point at $\Delta/2$, halfway between the source and the receiver (Fig. 3.5-6, *top*). Their travel time is thus the sum of the travel times for two legs

$$T(\Delta) = T(\Delta/2 + \epsilon) + T(\Delta/2 - \epsilon). \quad (1)$$

Using the first two terms of the Taylor series

$$\begin{aligned} T(\Delta/2 + \epsilon) &\approx T(\Delta/2) + \epsilon \frac{dT}{d\Delta} + \frac{\epsilon^2}{2} \frac{d^2T}{d\Delta^2}, \\ T(\Delta/2 - \epsilon) &\approx T(\Delta/2) - \epsilon \frac{dT}{d\Delta} + \frac{\epsilon^2}{2} \frac{d^2T}{d\Delta^2}, \end{aligned} \quad (2)$$

shows that

$$T(\Delta) \approx 2T(\Delta/2) + \epsilon^2 \frac{d^2T}{d\Delta^2}. \quad (3)$$

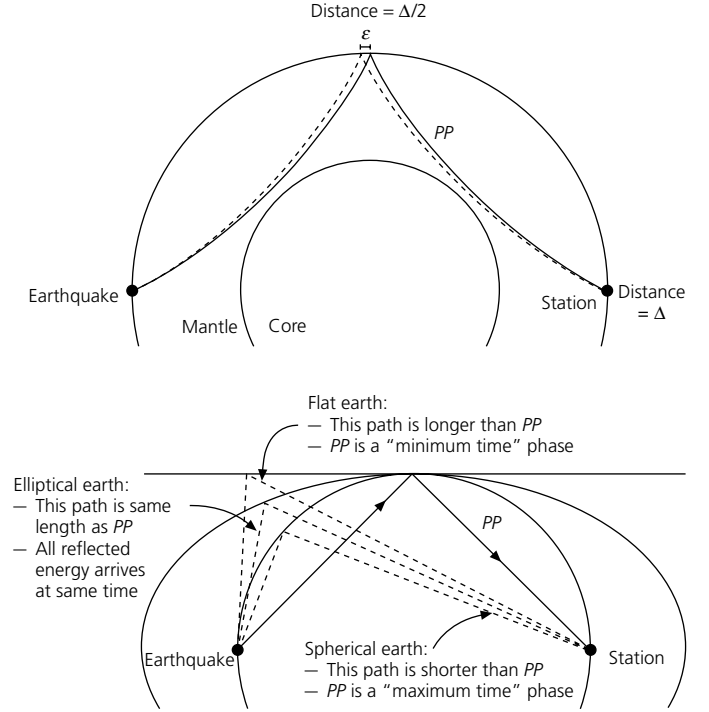


Fig. 3.5-6 *Top*: Ray path for a surface reflection. The reflection is a maximum-time phase, because the travel time for reflection at the midpoint $\Delta/2$ is longer than on nearby alternative paths. *Bottom*: Ray paths for a surface reflection in a homogeneous medium, in which all reflections off the elliptical surface have the same travel time. The reflection off the midpoint is a minimum-time phase if the surface is flat, and a maximum-time phase if the surface is circular.

By Fermat’s principle, the true ray path is that on which the derivative of travel time with respect to ϵ is zero,

$$\frac{dT}{d\epsilon} = 2\epsilon \frac{d^2T}{d\Delta^2} = 0, \quad (4)$$

so ϵ is zero, giving the expected bounce point. To see if this is a minimum or a maximum, we form the second derivative

$$\frac{d^2T}{d\epsilon^2} = 2 \frac{d^2T}{d\Delta^2}. \quad (5)$$

Figure 3.5-4 and Section 3.4.2 show that the direct P and S waves have travel time curves that are concave down, $d^2T/d\Delta^2 < 0$, so their surface reflections PP and SS are maximum-time phases. Thus PP or SS waves traveling along the same azimuth that reflect at the surface either closer or further than the point where PP reflects arrive earlier. By contrast, the core reflections like ScS have travel time curves that are concave upward, so in Eqn 5 $d^2T/d\Delta^2 > 0$, and its surface reflection ScS2 is a minimum-time phase.

An intuitive way to view minimum- and maximum-time phases is to consider ray paths for surface reflections in a homogeneous medium (Fig. 3.5-6, *bottom*). An ellipse defines the set of points whose summed distances to two points, or foci, are equal. Thus, if an earthquake and a receiver were the foci, the travel time for a reflection from any point on the ellipse would be the same. Hence, if the surface were elliptical, the reflected phase would be neither a minimum- nor a maximum-time phase, because all the energy would arrive at the same time. If the surface were flat, and thus had less curvature than the ellipse, waves that reflect off the surface slightly closer or further than the midpoint travel further, making the reflection a minimum-time phase. However, if the surface were circular and more curved than the ellipse, waves reflected off the surface slightly closer or further than the midpoint travel a shorter distance, making the reflection a maximum-time phase. This last case is analogous to that for *PP* and *SS* in the spherical earth.

Although *PP* and *SS* are maximum time phases with respect to distance, they are minimum travel time phases with respect to azimuth, as are most phases. Thus waves with a bounce point off the great circle path between the source and the receiver arrive later. This combination of maximum time with respect to distance and minimum time with respect to azimuth makes the surface reflections sample an “X”-shaped region of the surface, known as the *Fresnel zone* (Section 3.7.3), near the bounce point. The fact that these are maximum-time phases also causes them to undergo a $\pi/2$ phase shift³ (Fig. 2.6-5). Each successive bounce at the surface causes another $\pi/2$ phase shift, so *SSS* is phase-shifted by π and inverted with respect to direct *S*. *S4* undergoes a $3\pi/2$ phase shift, and *S5* has a 2π shift, giving it the same shape as the original *S*.

Figure 3.5-5 is drawn for an earthquake beneath the earth’s surface. Because earthquakes occur to depths of 700 km, seismic ray paths go up from earthquakes as well as down. Lower-case “*p*” and “*s*” identify upgoing compressional and shear waves (Fig. 3.5-2). *pP* goes up as a *P* wave and reflects near the epicenter, whereas *sP* goes up as an *S* wave and converts to a *P* wave at the surface. These reflections are useful because the travel time difference between direct *P* and *pP*, for example, indicates the depth of the earthquake. After an upgoing wave reflects at the free surface, it can undergo later conversions, so *pPP*, *sPS*, etc. are possible arrivals.

Many other body wave phases have been identified and are included in travel time tables. In addition, some tables give arrival times for Love and Rayleigh surface waves. As shown in Fig. 2.7-4, these surface waves are dispersive, so different frequencies have different arrival times, making the time shown approximate. This time is still useful for various purposes, including allowing earth structure studies to avoid phases that may be obscured by surface waves. In many cases, deep

earthquakes are used for body wave studies, because they generate only small surface waves.

Finally, it is worth remembering that travel time tables are compiled from observations of seismic arrivals. Although most arrivals on seismograms can be identified today from existing tables, important results are still found by noticing and explaining a previously unrecognized arrival.

3.5.2 Core phases

The contrast in properties between the solid mantle and the liquid core, which has lower velocity than the mantle above, makes the core well suited to seismological study using reflected, transmitted, converted, and diffracted arrivals.

Core reflections are of great interest because the core–mantle boundary (CMB) is a solid–liquid boundary, and thus a strong reflector for shear waves. Reflections off the CMB are denoted by a lower-case “*c*,” so *ScS* is an *S*-wave reflection and *PcP* is a *P*-wave reflection. Conversions at the CMB also occur. *ScP* goes down through the mantle as a shear wave and returns as a compressional wave, whereas *PcS* does the reverse. Some phases undergo multiple reflections at both the core and the surface; *ScSScS* (or *ScS2*) bounces twice at the CMB and once at the surface. Such reflections, known as multiple *ScS*, are shown in Fig. 1.1-4.

ScS is a more distinct arrival than *PcP*, because the liquid core does not transmit shear waves. The *SH* part of the motion in the incident *ScS* cannot convert to *P* waves at the CMB, so is totally reflected. Hence *ScS* is often well recorded on the transverse component (Section 2.4.4) of a seismometer. By contrast, the *PcP* reflection is generally weak, because the impedance contrast (Section 2.6.6) is small, so most *P* energy incident on the CMB is transmitted. The small impedance contrast (about 5%) arises because the *P*-wave velocity decrease going from the mantle to the core (about 13.7 km/s to 8.1 km/s) is offset by the density increase (about 5.5 g/cm³ to 9.9 g/cm³).

Core reflections, especially *ScS*, are useful in studies of earth structure, because they give a vertical average velocity for the mantle. The travel time curves for these phases are concave upward (Fig. 3.5-4), like that for the reflection off the top of a layer in a flat geometry (Section 3.2.1). Similarly, they have finite travel time at zero distance because of the time needed to get down to the core and back.

The travel times and amplitudes of core phases are also used to study structure near and within the core, because their ray paths are sensitive to the structure. To illustrate this idea, consider *P*-wave ray paths (Fig. 3.5-7, *top left*) within the earth. Rays leaving the source at progressively smaller angles of incidence (closer to the vertical) bottom deeper in the mantle and so reach greater distances. As the bottoming depth approaches the core–mantle boundary, the travel times of *P* and *PcP* converge (Figs 3.5-3 and 4). Eventually, at about 98° (the precise distance depends on the depth of the earthquake and the exact velocity structure), *P* grazes the core–mantle boundary, and *P* and *PcP* are identical.

³ This phase shift, also known as a Hilbert transform, can be viewed by thinking of the pulse as made up of sine and cosine functions and turning each into the other.

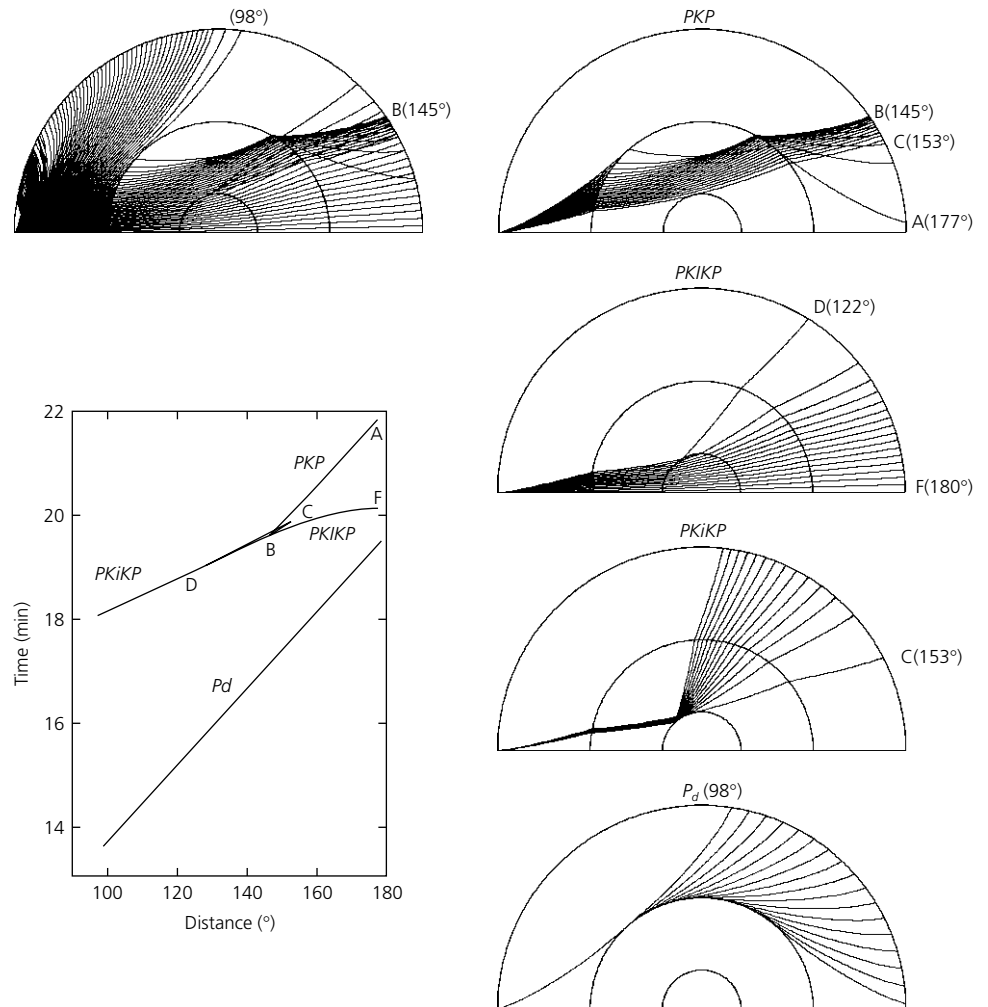


Fig. 3.5-7 Ray paths and travel times for major core phases, computed for earth model PREM. *Top left:* Paths for direct rays (i.e., excluding reflections and diffractions). *Right:* Ray paths for four other phases: *PKP* passes through the outer core, *PKIKP* also penetrates the inner core, *PKiKP* reflects from the boundary between the outer and inner cores, and *P_d* (also called *P_{diff}*) diffracts along the core–mantle boundary. *Lower left:* Travel time curves for these phases. Points on the earth’s surface are labeled with their distances in degrees.

A ray with a slightly smaller angle of incidence, however, refracts downward at the CMB, because the core has a lower *P* velocity than the mantle. It thus enters the core, travels through it, refracts into the mantle, and reaches the surface. This phase is called *PKP*, where “*K*” denotes passage through the outer core.⁴ For an angle of incidence slightly below grazing, *PKP* reaches the surface at point A (Fig. 3.5-7, *top right*), at a distance close to 180°. Rays with smaller angles of incidence penetrate deeper into the core, and thus arrive at distances successively less than 180°, down to a distance of about 145° (point B). At this point the pattern reverses, because rays with smaller angles of incidence arrive at successively greater distances. This goes on for rays reaching distances up to point C (~153°, depending on the earth model), corresponding to the ray that grazes the inner core–outer core boundary.

The ray paths show that the low velocity in the outer core gives rise to a geometrical shadow zone, where Snell’s law

predicts that no direct rays arrive.⁵ We have seen (Fig. 3.4-7) that the corresponding travel time curve should have a break due to the shadow zone, and then two branches on the far side of the shadow zone. For the core, the shadow zone occurs for distances between ~98° to ~145° (point B, Fig. 3.5-7, *top left*). Beyond 145°, the travel time curve has two branches for *PKP*. The AB branch (sometimes labeled *PKP₂*) is the back branch, on which rays with smaller angles of incidence appear at smaller distances, whereas the BC branch is the forward branch on which rays with smaller angles of incidence appear at larger distances.

In reality, body waves are observed in the shadow zone. Much of the body wave energy arrives as surface-reflected (*PP*, *PPP*, *SS*, etc.) or multiply core-reflected (*ScS2*, etc.) arrivals. Other arrivals are due to *P* waves that encounter the inner core. Because the inner core has higher *P*-wave velocity than the

⁴ “*K*” is from *Kern*, the German word for core.

⁵ Although the core’s existence had been inferred from the earth’s gravity (Section 3.8), the discovery of this shadow zone in 1906 by Richard Oldham (1858–1936) provided the first direct evidence and set the paradigm for future core studies.

outer core, waves refract upward and emerge in the shadow zone. These phases are known as *PKIKP*, because *P* waves in the inner core are denoted by “*I*.” In addition, waves reflect at the boundary between the inner and outer cores, giving the phase *PKiKP*.⁶ (The lower-case “*i*” is analogous to the lower-case “*c*” in *PcP*.) The travel time curve thus has a *PKIKP* branch *DF*, where *D* is the distance at which *PKIKP* is first observed, and a back branch for *PKiKP*. The back branch begins at *C*, where *PKiKP* and *PKP* are the same, and extends through *D* back to zero distance (Figs 3.5-3 and 4), because the reflection occurs at vertical incidence. Hence the portion of the travel time curve containing *CD* and *DF* is due to the rapid increase of velocity at the inner core–outer core boundary, and is analogous to a triplication.

Seismic energy also enters the shadow zone via *P* and *S* waves that diffract around the core (Section 2.5.10). The ray paths for the diffracted *P* waves (denoted *P_d* or *P_{diff}*) shown in Fig. 3.5-7 represent energy that diffracted around the core, left the CMB, and traveled back to the surface. This process is much like that discussed for the head wave (Section 3.2.1). Thus, once the direct *P* wave becomes the diffracted wave at a distance near 100°, its travel time curve (Fig. 3.5-4) loses the curvature it had, because successive rays penetrated deeper to higher-velocity material. Instead, it becomes linear because all the diffracted waves bottom at the CMB, and so have the same ray parameter and hence apparent velocity. As for the head wave, assuming that the energy followed a ray path gives the diffracted wave’s travel time but cannot fully describe its amplitude, because diffraction involves energy propagating as waves, not rays. However, we will see that more complete formulations such as normal modes predict both the times and the amplitudes of the diffracted phases.

Figure 3.5-7 shows that the travel time curve for the core phases is complicated because it combines the effects of a geometric shadow zone, which gives two *PKP* branches, a triplication-like feature containing the *PKIKP* and *PKiKP* branches, and a diffraction branch. In reality, even these models are simplifications of a more complex reality. Figure 3.5-8, showing the travel times of several million *PKP* arrivals, illustrates several significant deviations from the theoretical curves in Fig. 3.5-7. First, the arrivals do not fall along narrow lines. This is partly due to errors of observation, but also due to the heterogeneous structure of the crust, mantle, and core, which makes some arrivals early and others late. Second, the *PKP-BC* branch continues beyond its geometrically predicted limit of 153°. This is because the *PKP-BC* wave diffracts around the inner core, although its amplitude decreases rapidly in the process, so there are few observations beyond 160°.

Third, and most importantly, the *PKP* travel times show an additional branch not predicted by geometric ray theory. These arrivals, labeled *PKP precursors*, appear to be a continuation of the *PKP-AB* branch and arrive as much as 20 s before the

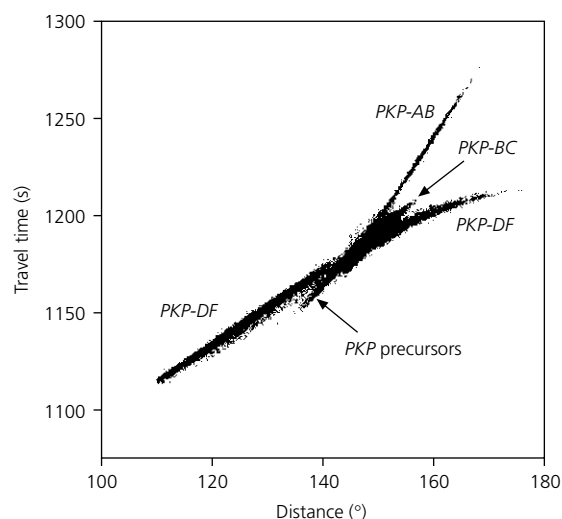


Fig. 3.5-8 Arrival times of *PKP* waves recorded by the International Seismological Centre during 1964–87. A point is plotted if there are at least 200 arrivals in the catalog for that time and distance. Although these arrival times are similar to the predicted travel time curves in Fig. 3.5-7, there are some differences. The *PKP-BC* branch is observed beyond its geometrical limit (153°) due to diffraction around the inner core, and precursors to the *PKP-DF* branch are observed that result from seismic scattering at the CMB and in the mantle. (Courtesy of K. Koper.)

PKP-DF branch. These arrivals puzzled seismologists until it was realized that they were waves reflected, or scattered, from inhomogeneous structures in the mantle. This scattering is analogous to that discussed in Section 3.3.7 in the context of migration in reflection seismology. Because the scatterers are comparable in size (about 10–15 km) to the wavelengths of short-period *P* waves in the lower mantle, they behave as Huygens’ sources (Section 2.5.10). Thus a *PKP-AB* wave interacting with a scatterer at the CMB radiates waves in all directions (Fig. 3.5-9). Those arriving before *PKP-DF* are clearly observed, whereas those arriving afterwards are lost amid *PKP-DF*. The range of observable scattered *PKP* waves is shown as the shaded regions in Fig. 3.5-9, illustrating another way in which seismic energy reaches the shadow zone. Although most such scattering occurs near the CMB, modeling of the *PKP* precursors suggest that waves are also scattered by small reflectors throughout the mantle, as shown by the dark shaded region in Fig. 3.5-9.

Some core phases begin as *S* waves (Fig. 3.5-5). Although no *S* waves propagate in the liquid outer core, phases like *SKS* travel through the mantle as an *S* wave and through the core as a *P* wave. *SKKS* is similar to *SKS*, but also involves an underside reflection at the CMB. Because the *P* velocity of the uppermost core (about 8.1 km/s) is not much larger than the *S* velocity of the lowermost mantle (about 7.2 km/s), *SKS* and *SKKS* waves do not change direction significantly as they cross the CMB. Thus *SKS*, *SKKS*, *SKKKS*, etc. are the only waves that bottom near the top of the core and are used to constrain the outer core’s velocity structure.

⁶ Observations of this phase by Inge Lehmann (1888–1993) in 1936 provided the first evidence for the existence of the inner core.

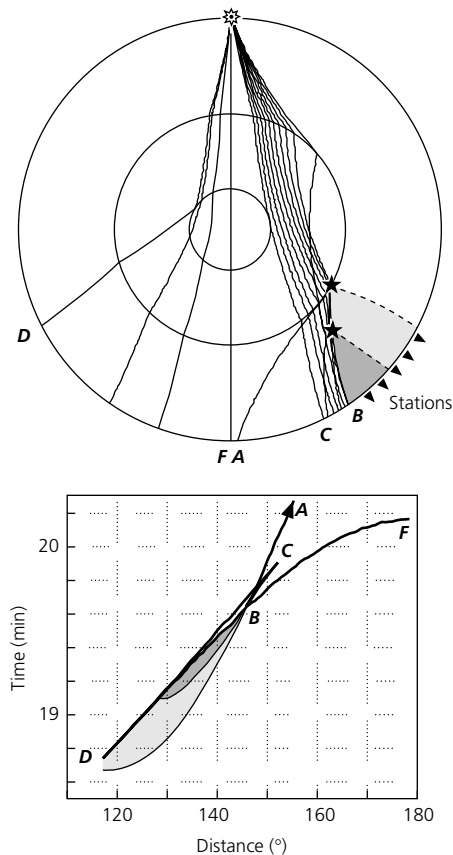


Fig. 3.5-9 A model for the PKP precursors shown in Fig. 3.5-8. *Top*: PKP-AB waves that interact with scatterers (stars) cause arrivals at distances less than the geometrically allowed AB range. As shown by the travel times (*bottom*), these arrivals precede the PKP-DF arrivals. Scatterers at the base of the mantle yield waves in the light and dark shaded regions, and others in the mid-mantle yield waves in the dark shaded region. (Hedlin *et al.*, 1997. Reproduced with permission from *Nature*.)

Other core phases, some of which are not included in the travel time plots in Fig. 3.5-4, have also been reported. These include PKKP (Fig. 3.5-10), a P wave that has undergone underside reflection at the CMB, PKPPKP (sometimes called P'P'), a PKP phase reflected at the surface, and PKIIKP, an underside reflection from the outer core–inner core boundary. An especially elusive phase has been PKJKP, which, by analogy to PKIKP, travels through the inner core as an S wave. The weak amplitude of this phase, combined with the fact that it arrives late in the seismogram amid other phases, has made it difficult to observe. PKJKP has been verified only recently, by stacking data from very large deep earthquakes that generate the large body waves needed to produce even small PKJKP, while not generating surface waves that mask the small core arrivals.⁷

⁷ This observation and normal mode results are overcoming seismologists' prior reservations, exemplified by comments like "the inner core, which may exist, is said to have the following properties. . ."

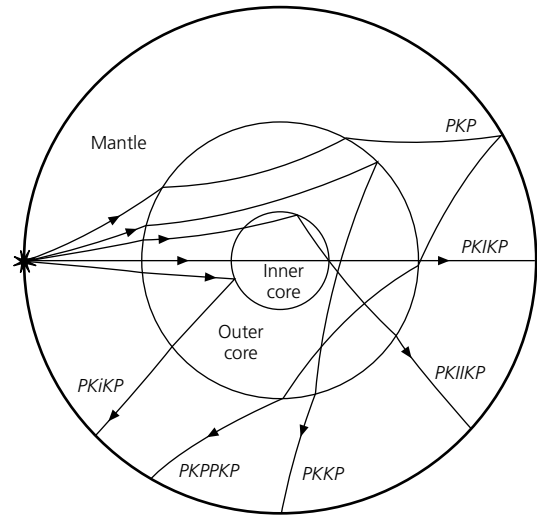


Fig. 3.5-10 Some additional core phases. PKKP and PKIIKP are underside reflections at the core–mantle boundary and outer core–inner core boundaries, and PKPPKP (P'P') is an underside surface reflection.

Core phases can be challenging to study with travel time data because their travel time curves are complicated and some of the arrivals are small. Amplitude and waveform studies provide additional information. As we have seen (Section 3.2.3), amplitudes can be used to differentiate between structures that would give similar travel times. Some insight into the amplitudes can be obtained from the ray densities (Section 3.4.2). For example, the AB and BC branches of the PKP travel time curve meet at the far side of the shadow zone, at point B. Figure 3.5-7 shows that rays which left the source at uniform angle increments are concentrated there, so large amplitudes are expected at this caustic.

This discussion of amplitudes brings out another interesting point. Although the earth is approximately spherical, we have discussed only waves propagating in the plane containing the source, the receiver, and the center of the earth. One case in which sphericity is important is near the antipode, the point 180° from the source. Figure 3.5-11 shows seismograms recorded at PTO (Porto, Portugal) and MAL (Malaga, Spain) from an earthquake in New Zealand. Phases like PP and PKP are focused at the antipode, because paths in any direction from the source arrive at the same time. Note the larger arrivals at PTO, only 0.7° from the antipode.

3.5.3 Upper mantle structure

The velocity structure of the upper mantle shows two major effects. First, it has discontinuities and velocity gradients that are essentially radially symmetric, which are believed due to the effects of pressure on the minerals present. Second, it contains significant lateral heterogeneity that is primarily associated with temperature variations due to cold subducting oceanic lithosphere. We discuss the radial velocity structure here,

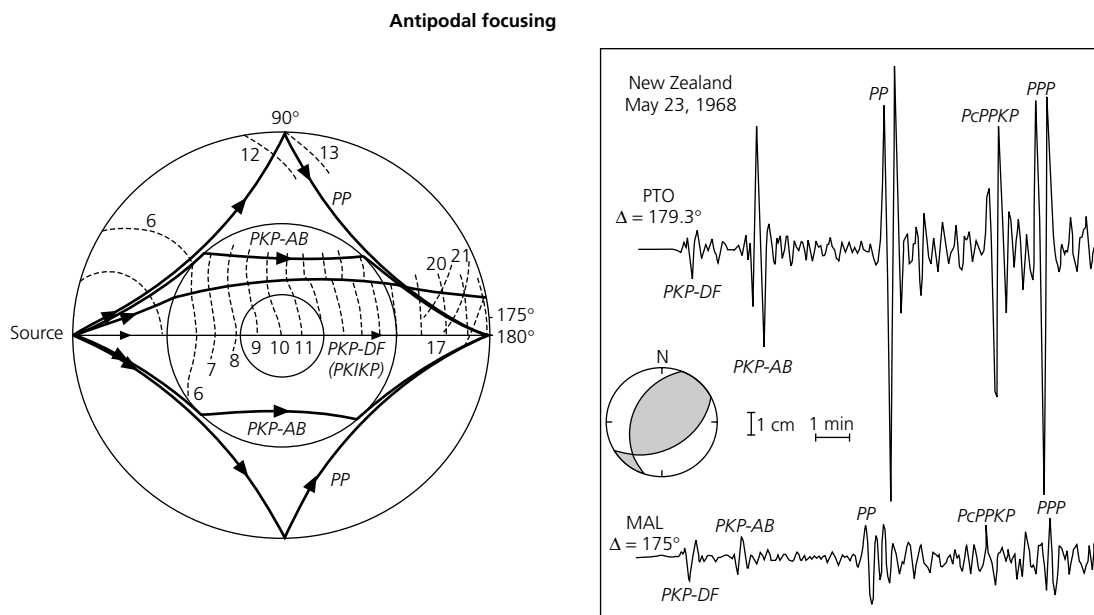


Fig. 3.5-11 Focusing of *P* waves at the antipode, 180° from an earthquake. *Left*: Seismic rays *PKP-AB* and *PP* focus at the antipode. Dashed lines represent wave fronts, whose propagation time in minutes is given. *Right*: Seismograms showing antipodal focusing of core phases. (Rial and Cormier, 1980. *J. Geophys. Res.*, 85, 2661–8, copyright by the American Geophysical Union.)

explore its mineralogical causes in Section 3.8, and consider the effects of subducting lithosphere in Section 5.4.

We have already discussed the velocity structure of the uppermost mantle shown by surface wave dispersion (Section 2.8). Body wave analyses reveal a similar structure. The sub-crustal lithosphere shows generally fast *P*- and *S*-wave velocities of about 8.1 and 4.5 km/s. This high-velocity layer provides a way of defining the lithosphere, termed the *seismic lithosphere* or *lid*, from seismological observations. The thickness of the seismic lithosphere varies with location. At mid-ocean ridges, where oceanic plates are created, its thickness approaches zero. Beneath stable cratons, the fast lithospheric velocities extend to about 200 km. As a global average, the seismic lithosphere extends to about 80–100 km depth.

In most regions of the world, we find a seismic low-velocity zone (LVZ) beneath the seismic lithosphere. The LVZ approximately coincides with the expected mechanically weak asthenosphere underlying the stronger lithosphere. The lithosphere and asthenosphere are defined by their mechanical properties, such that plates of strong lithosphere slide over weaker asthenosphere. This contrast, as we will see, results from the fact that the lithosphere is the cold outer thermal boundary layer of the solid earth (Sections 3.8, 5.1). By contrast, the high-velocity seismic lithosphere and underlying LVZ are seismologically defined entities. The rough correspondence between the seismological and mechanical layers indicates that the two are closely related, and that seismic observations can be used to map mechanical structure. The two sets of layers are not identical for several reasons including the fact that seismic

waves sample physical properties over a period of seconds, whereas the lithosphere and asthenosphere are inferred from data sampling periods of thousands and millions of years (Section 5.7).

The depth and magnitude of the LVZ vary regionally. In tectonically active regions like western North America, the LVZ is well developed and relatively shallow. In stable continental regions that have not experienced tectonism for a long time, the LVZ is deeper and less pronounced, and may not even be present. The thick, high-velocity layer under continents has led to the suggestion that it may reflect a chemically distinct *tectosphere*. For this hypothesis, continents behave differently from the oceanic lithosphere, where surface wave dispersion shows a pronounced LVZ for all ages (Fig. 2.8-7). This persistence may reflect the fact that oceanic lithosphere is never older than 180 Ma, tectonically young by continental standards, because older oceanic lithosphere is subducted away.

We will show in Section 5.7 that the contrast between the high-velocity seismic lithosphere and the asthenosphere LVZ is probably related to variations in material strength between the cold lithosphere and the warmer asthenosphere. There may also be some effects of partial melting. This situation differs from the velocity differences between the crust and the mantle, which result from their different compositions. Beneath the LVZ, which extends to an average depth of about 200 km, temperatures increase only slowly, but velocities increase significantly in response to the increasing pressure.

The transition zone between the upper and lower mantles is marked by the velocity discontinuities at depths of about 410

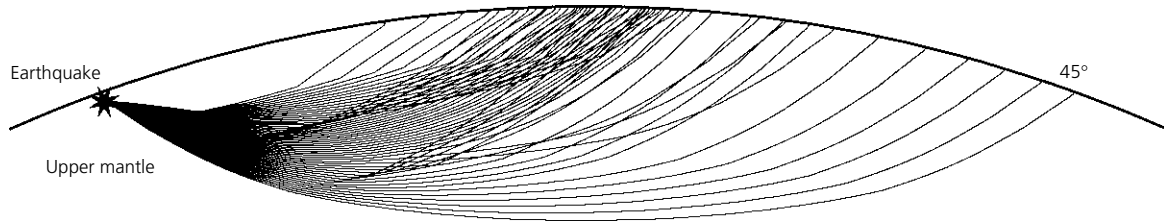


Fig. 3.5-12 Ray paths for P waves through the upper mantle, computed for earth model PREM, showing triplications due to mantle discontinuities.

and 660 km. We saw in the last section that a rapid velocity increase (Fig. 3.4-6) produces a triplication in the travel time curve. Upper mantle travel times show two triplications around 15° and 22° caused by the 410 and 660 km discontinuities. Ray paths for such a structure are shown in Fig. 3.5-12. Some reference models such as PREM also have a discontinuity at 220 km, and regional studies also often find discontinuities at other depths in the upper mantle.

One difficulty in studying the transition zone, or other regions of complex velocity structure, is that travel time curves are composites of data from many earthquakes at different distances. The process of combining the data can make the details of the triplication difficult to observe. Moreover, $dT/d\Delta$, the derivative of the travel time curve that is used in inverting for velocity, is uncertain due to the scattered data. These difficulties can be addressed in several ways. One is to derive information from the waveforms as well as the travel times. A second is to use arrays of seismometers spaced closely enough that it is possible to identify arrivals corresponding to the different branches of triplications and directly measure $dT/d\Delta$ by tracing them across the array. Such dense data also facilitate waveform studies.

Figure 3.5-13 (overleaf) illustrates these ideas with an array study of upper mantle P -wave structure under the Gulf of California spreading center. Data from ten earthquakes are combined into a record section for the epicentral distance range $9\text{--}40^\circ$. The travel time curves show two triplications, one near 15° due to the 410 km discontinuity, and another around 22° due to the 660 km discontinuity. Travel times and synthetic seismograms predicted by the velocity structure (GCA) derived from the data fit the data well, including the back branches (C-B and D-E) of the triplications. The effects of the discontinuities appear in the $p(\Delta)$ data as two groups of later arrivals for which p increases with Δ . These arrivals are the back branches of the triplications (Fig. 3.4-6). The remaining arrivals show p decreasing with Δ , and thus are the forward branches.

Figure 3.5-14 compares the GCA model to upper mantle models for other tectonic environments: ARC-TR (arc-trench) for the Japan subduction zone, T7 for the tectonically active western portion of North America, and K8 for the stable Eurasian shield. Above 200 km, all show a LVZ overlain by a higher-velocity lid, but the depth and extent of the LVZs differ. The shield model, for example, has the thickest lid. Below 200 km, GCA shows the lowest velocity. The depths of

the 410 and 660 km discontinuities differ between the models. These differences are thought to reflect the fact that the mineral phase transformations causing the discontinuities occur at pressures (and hence depths) that depend on temperature. Thus lateral temperature changes, especially those associated with subduction zones, should change the depths at which these transitions occur (Section 5.4.2).

Waveform modeling provides additional information about the transition zone. For example, waveform modeling of intermediate-period S waves shows a discontinuity at about 520 km depth that is not observed with short-period P waves. The phase transition thought to cause this discontinuity may occur over a greater depth range than for the 410 and 660 km discontinuities, making it visible only to longer-period waves.

3.5.4 Lower mantle structure

Velocities increase rapidly with depth for roughly 100 km beneath the 660 km discontinuity, but then increase more slowly. The rapid increase implies that mineral transformations continue, whereas the slow increase implies that the mineralogy and composition of the material are not changing significantly, and that the velocity increases are primarily due to the material being compressed by higher pressure. However, weak seismic discontinuities have been reported at a variety of depths such as 900 and 1300 km. These may represent either global discontinuities like the 410 and 660 ones, or local velocity anomalies, perhaps due to fragments of old subducted slabs.

The situation changes dramatically in the D'' layer at the very base of the mantle, a fascinating and poorly understood region⁸ that has a velocity structure whose complexity rivals that of the lithosphere. D'' , the bottom few hundred kilometers of the mantle, was initially differentiated from the rest of the mantle (D') because the velocity gradient with depth is lower. This lower gradient is expected, because D'' is a thermal boundary layer between the mantle and hotter core. The expected $\sim 1000^\circ\text{C}$ temperature difference across D'' would lower velocities and thus decrease the velocity gradient.

However, detailed velocity models show that at the top of this lower-gradient region the velocity increases sharply

⁸ The uncertainties about D'' have been illustrated by describing its thickness as 250 ± 250 km (Jeanloz, 1990).

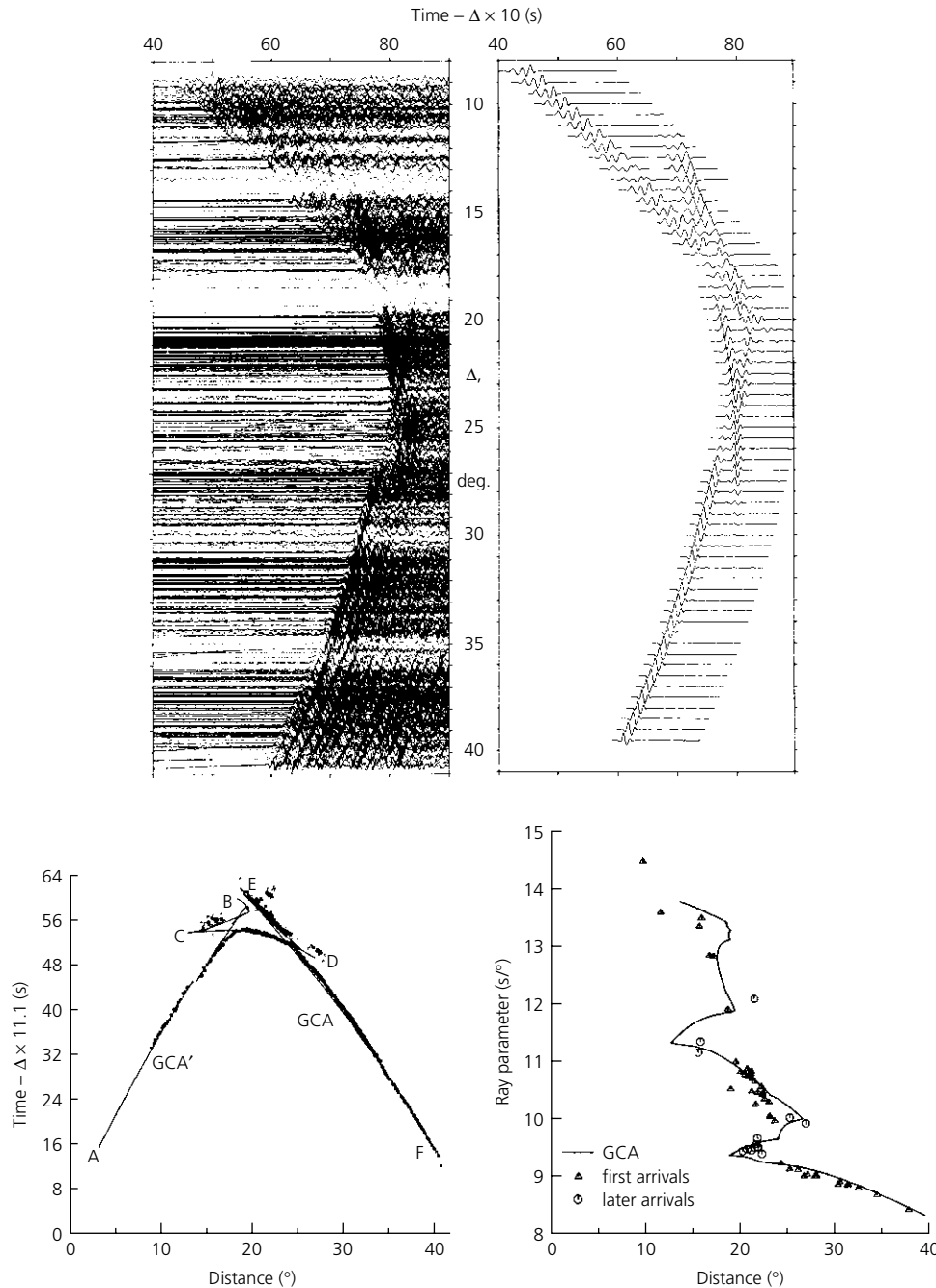


Fig. 3.5-13 Seismic array study of upper mantle structure. *Top*: Record sections, plotted with a reducing velocity of $10^\circ/\text{s}$, showing observed (*left*) and synthetic (*right*) seismograms. *Bottom left*: Reduced travel time plot, showing travel time data and model predictions. *Bottom right*: $p(\Delta)$ plot and model predictions. The two triplications are evident in the record sections, travel time plot, and $p(\Delta)$ plots. The slight break in the travel time curve at 13° is due to use of slightly different models (GCA' versus GCA). (Walck, 1984.)

(Fig. 3.5-15). A feature of such models is that the high- and low-velocity regions trade off to give similar travel times as PREM, which does not contain the high-velocity region. Thus D'' is now often delineated by the location of the discontinuous velocity increase, which averages about 250 km above the CMB. This is ironic, in that D'' was first named for a region of lower than expected velocities.

Observations of the velocity increase, known as the D'' discontinuity, are usually made with the phases PdP and SdS , each of which combines waves that reflect off and refract just under

the discontinuity (Fig. 3.5-16). PdP and SdS arrive between the direct (P and S) and core-reflected (PcP and ScS) phases, as shown. The discontinuity has been observed at many locations on the CMB, but other locations, even nearby, do not show a PdP or SdS arrival. Moreover, although the average depth of the discontinuity is 250 km above the CMB, the observed depths range from 100 to 450 km above the CMB.

One possible explanation for this variability is that the discontinuity has large topographic variations over small spatial wavelengths that focus and defocus waves. Another possibility

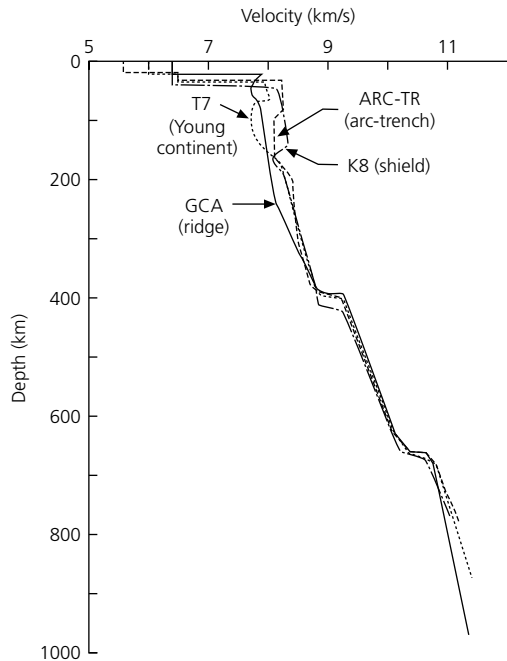


Fig. 3.5-14 Comparison of model GCA, derived from the data in Fig. 3.5-13, to P -wave velocity models for other tectonic regions. (Walck, 1984.)

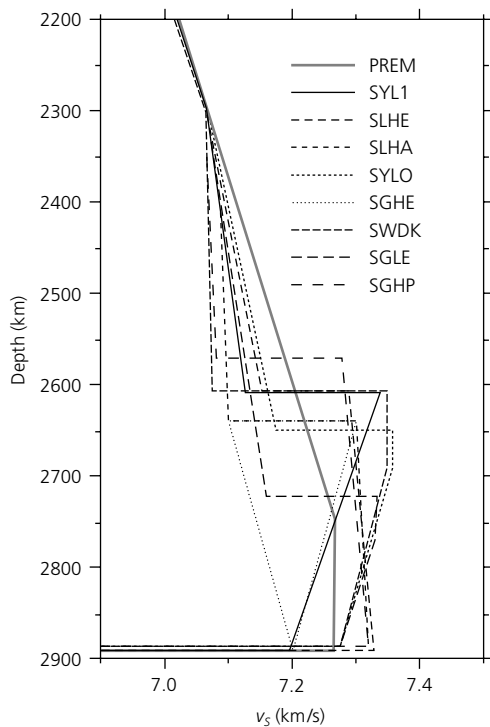


Fig. 3.5-15 Velocity structures from several studies showing an increase in velocity about 250 km above the core–mantle boundary, known as the D'' discontinuity. (Wyssession *et al.*, 1998. *The Core–Mantle Boundary Region*, 273–97, copyright by the American Geophysical Union.)

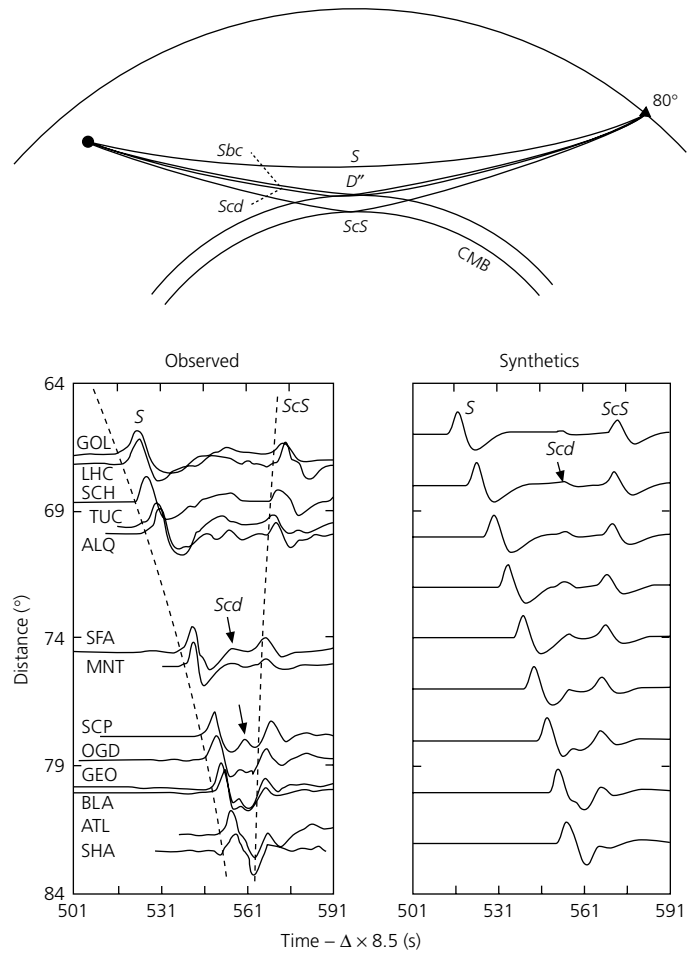


Fig. 3.5-16 Top: Schematic ray paths of the two arrivals making up the phase SdS . Sbc reflects off the D'' discontinuity, and Scd refracts just below it. (Wyssession *et al.*, 1998. *The Core–Mantle Boundary Region*, 273–97, copyright by the American Geophysical Union). Bottom: Observed SdS arrivals (arrows) (left) compared to synthetic seismograms (right) computed using velocity model SYLO (Fig. 3.5-15). (After Young and Lay, 1990. *J. Geophys. Res.*, 95, 17, 385–402, copyright by the American Geophysical Union.)

is that there is no actual discontinuity, but that complex three-dimensional velocity heterogeneities give the appearance of a discontinuity. This possibility is supported by observations of the increased scattering of seismic waves passing through D'' . In either case, it is possible that the increase in velocity is associated with subducted lithosphere that sank to the bottom of the mantle. There is a correlation between regions of fast velocities in D'' and the projected locations of fossil slabs from ancient subduction zones (Fig. 3.5-17), which should retain a cold thermal anomaly for a long time after reaching the CMB (Section 5.4.1). Seismic modeling suggests that this mechanism could generate PdP and SdS phases.

D'' shows additional complexities. There is strong evidence for significant seismic anisotropy (Section 3.6.6). Large lateral

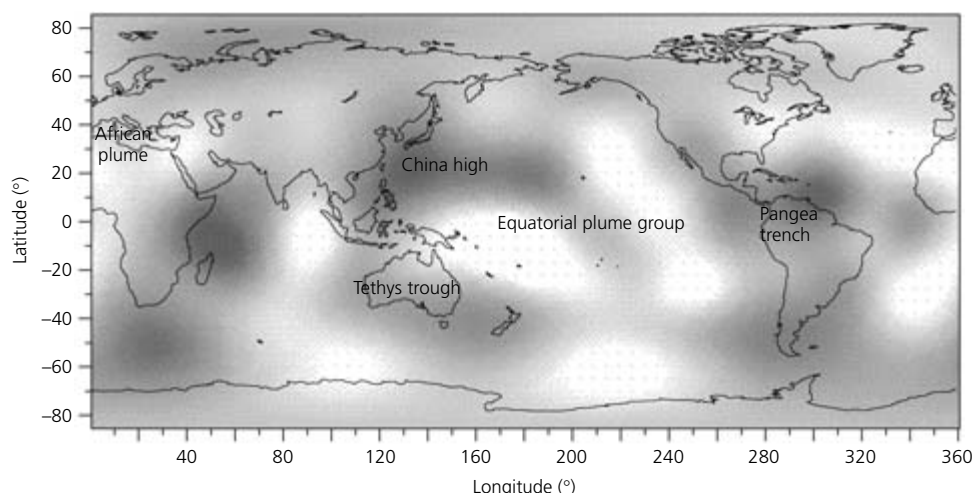


Fig. 3.5-17 *P*-velocity variations at the base of the mantle. Dark areas represent anomalously fast velocities, and light areas are slow anomalies. The fast anomalies correlate with the predicted locations of lithosphere subducted during the Mesozoic that sank to the base of the mantle. (Wyssession, 1996b. Reproduced with permission from *Nature*.)

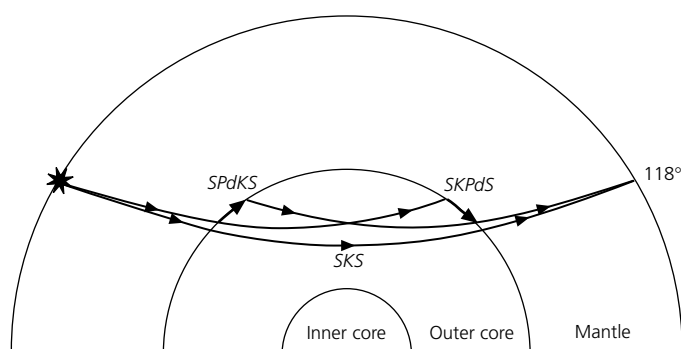


Fig. 3.5-18 Ray paths of *SPdKS*, a phase that is highly sensitive to the ultra-low-velocity zone at the base of the mantle. As with many studies of the deep mantle and core, it is analyzed using the difference between its travel time and another phase — in this case, *SKS*.

variations at both small and large spatial wavelengths occur for velocities within D'' and for topography on the CMB. There is also evidence for an ultra-low-velocity zone (ULVZ) at the very bottom 10–20 km of the mantle. The ULVZ is observed with an unusual body wave phase, *SPdKS*, which is similar to *SKS* but travels partly as a diffracted *P* wave at either or both of its entrance and exit points from the core. *SPdKS* appears as a shoulder of the *SKS* arrival, and is very sensitive to the *P*-wave velocity structure just above the CMB (Fig. 3.5-18). Modeling of *SPdKS* waveforms suggests that v_p may be 10% lower than in the rest of D'' , and the reflection coefficients of *PcP* precursors that reflect off the top of the ULVZ suggest that v_s may decrease by 30%. The ULVZ may result from partial melt, because it is most prominent where D'' velocities are slowest, implying that the high temperatures causing the low velocities may also cause more partial melting.

In summary, much uncertainty remains about the detailed structure of D'' and its causes. This is hardly surprising, because

the CMB is likely to be the site of many processes involving lateral and vertical motions and vigorous chemical reactions. An analogy might be that D'' is a thermal boundary between the mantle and the core, analogous to the lithosphere, which is the thermal boundary layer at the top of the mantle. The high-velocity layer at the base of D'' may be a chemical layer, analogous to the crust. These complexities have led the CMB to be called the graveyard of ancient ocean lithosphere, the birthplace of mantle plumes, and the region that most significantly controls the outer core convection patterns and thus the earth's magnetic field. The fact that we study this region largely via seismic “remote sensing” through 2890 km of heterogeneous mantle may limit the degree to which it can be understood.⁹

3.5.5 Visualizing body waves

To end our discussion of body waves, it is worth considering their physical nature. We have treated body wave arrivals like *S* and *ScS* as geometric rays. However, although it is convenient to describe these waves as rays and to show their paths through ray tracing, this approximation does not fully describe their behavior.

To see this, we consider a numerical simulation showing time snapshots of the *SH* shear wave field generated by a 600 km-deep earthquake (Fig. 3.5-19). The wave field is synthesized by summing 28,000 torsional normal modes (Section 2.8) with periods above 12 s. The calculations show accurate relative amplitudes, with light and dark shades representing displacements into and out of the paper, respectively. Although the normal mode solution is itself an approximation to the actual wave field in the laterally heterogeneous earth, it is much closer to reality than geometric rays.

⁹ The geophysical significance of the CMB and the large uncertainties remaining about it are summarized by D. Stevenson's description of D'' as “the sum of all of our ignorance of the interior of the earth.”

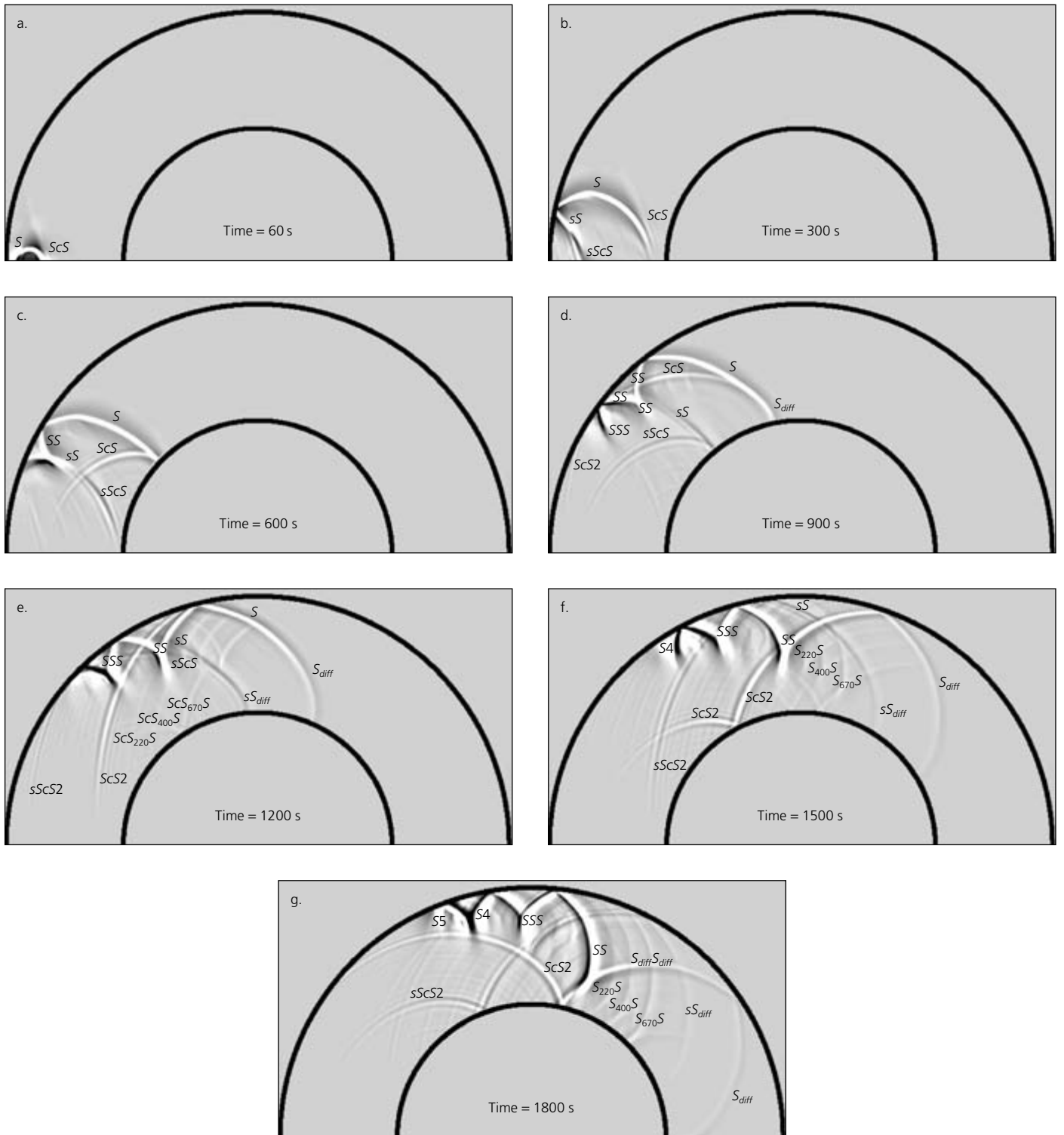


Fig. 3.5-19 Snapshots of a synthetic *SH* wave field showing the propagation of waves after a 600 km-deep earthquake. The initial wave front moves away from the source at the lower left side of the figures. The wave front develops complexity due to interactions with the surface, CMB, and internal discontinuities and velocity gradients. The wave field is computed using the spherically symmetric PREM velocity model. Amplitudes are raised to a power of 0.8 to enhance smaller signals. (After Wyssession and Shore, 1994.)

As shown, a single spherical S wave front is quickly broken into various wave fronts by reflections off the surface, mantle discontinuities, and the core. As the wave fronts arrive at the surface, they cause arrivals that we call S , ScS , sS , etc.

In the first frame, Fig. 3.5-19a, which is 60 s after the earthquake, the wave front maintains much of its initially spherical shape. The upgoing part of the wave front is headed toward the surface, but will not reach it for another 67 s. The downgoing part of the wave front is headed toward the core, where it will be fully reflected and give rise to ScS .

In Fig. 3.5-19b, 300 s after the earthquake, the wave front still maintains its integrity, though the upper part is now reflecting off the surface, and the lower part is about to reach the core. The slower upper mantle velocities cause bends in both the reflected and the unreflected waves. The S wave front is reaching the surface 12.5° away from the source, and at closer distances has already reflected downward. When these downgoing waves reach the surface again, they will be called the sS and $sScS$ phases. The downgoing portion of the initial wave front that will become ScS has not yet reached the core.

By 600 s after the earthquake (Fig. 3.5-19c) added complexity is evident. The upgoing wave that reflected at the core will generate ScS and its multiples ($ScS2$, $ScS3$, etc.). The surface-reflected wave is separating into two parts. One is heading into the lower mantle and will eventually reach the surface as the $sScS$ and sS phases. The other will turn higher up in the mantle and arrive at the surface as the SS phase. Behind the sS , ScS , and $sScS$ wave fronts are upper mantle echoes reflected from the 220, 400, and 670 km discontinuities. However, despite all that is going on, the only phase yet recorded at the surface is S , now arriving 31° away from the source. sS will begin to arrive in another 63 s, at a distance of 24° .

By 900 s after the origin time (Fig. 3.5-19d), four segments of the broken wave front are reaching the surface: S at 52° , sS at 39° , SS at 38° , and ScS at 33° . The sS and SS wave fronts have begun to separate. In contrast, the ScS and S wave fronts have begun to come back together because they enter the core shadow, where the S/ScS wave front continues as a diffracted S_{diff} wave. Behind the S and ScS waves in the lower mantle are the sS and $sScS$ waves, which follow similar paths except for their surface reflections. The distance between S and sS (and also between ScS and $sScS$) is a function of the depth of the earthquake. The three wave front segments labeled SS form a characteristic “Y” shape that results from the waves turning in the mid-mantle. The “Y”’s junction represents the superposition of the part of the wave front that is heading down toward the bottoming point and the part of the wave front that has already turned and is heading back up again. Behind SS , the phase SSS that bounces twice on the underside of the surface, is beginning to form.

In Fig. 3.5-19e, 1200 s after the earthquake, most of the initial S wave front is actually S_{diff} because S grazes the core at about 100° . The surface-reflected sS wave now also diffracts around the core as sS_{diff} . SSS is now fully developed, and is reaching the surface behind SS . The polarity of SSS is different

from that of SS , because each successive surface bounce changes its phase by $\pi/2$ (Section 3.5.1). The initial S -wave polarity is into the page (light-colored), whereas SSS is primarily out of the page (dark colored) because it has been phase-shifted twice. The smaller-amplitude phases evident are reflections from the upper mantle discontinuities in the velocity model at depths of 220, 400, and 670 km, and so come in threes. One set of these, labeled as $ScS_{220}S$, $ScS_{400}S$, and $ScS_{670}S$, are underside reflections that precede $ScS2$.

By 1500 s after the earthquake (Fig. 3.5-19f), the initial wave front is entirely diffracted S_{diff} , reaching the surface at a distance of 111° . Because waves travel much faster at the base of the mantle than in the upper mantle, S_{diff} at the CMB has gone further, reaching 152° . A set of mid-mantle reflections labeled $S_{220}S$, $S_{400}S$, and $S_{670}S$, which are also visible in the previous panel, appear ahead of SS . These peel off the upgoing S/S_{diff} wave front as it interacts with the discontinuities. Because they are related to SS , they also have the “Y”-shape characteristic of underside-reflected phases. The upgoing parts of the “Y” formed from the upgoing S phase, but the downgoing parts (right side of the “Y”) peel off S_{diff} and are better called $S_{diff200}S_{diff}$, $S_{diff400}S_{diff}$, and $S_{diff670}S_{diff}$. The waves with the largest amplitudes, SS and SSS , are arriving at the surface at distances of 76° and 63° .

In Fig. 3.5-19g, 1800 s after the earthquake, $S4$ has begun to be observed at the surface (71°), following SS (97°) and SSS (83°). The next surface reflection, $S5$, is now developing. The $ScS2$ multiple reflection is arriving at the surface 36° from the earthquake. The downgoing part of SS is from S_{diff} reflecting at the surface, so it will arrive at the surface at distances greater than 200° as the phase $S_{diff}S_{diff}$. By now, 30 minutes after the earthquake, seismic energy has spread throughout the mantle. Multiple ScS waves are still reverberating between the surface and the core. At the CMB, the leading S_{diff} wave has wrapped around the antipode and is heading back toward the epicenter.

This simulation illustrates that although the ray paths used to describe body waves in the earth are intuitively appealing and useful, they are simple ways of characterizing a complicated wave field. An earthquake generates an initially spherical wave front whose interaction with various interfaces gives rise to many wave fronts. We use names for the arrivals that the wave fronts cause at the surface, so different parts of the same wave front, or the same part at different times, are given different names. Hence our intuition based on geometric rays can lead us to miss some of the richness that occurs. For example, we tend to view diffraction as an exotic effect different from the direct ray path, but the simulation shows no major change as the direct wave becomes the diffracted wave, although there is a loss of high frequencies. Hence the simulation shows no obvious core shadow zone, because seismic energy reaches the shadow zone by diffraction and multiple reflections. The essential point is that the wave fields are the physical entities, whereas rays are useful approximations whose limitations should be kept in mind.

3.6 Anisotropic earth structure

3.6.1 General considerations

So far in this chapter, we have considered a view of the earth developed from analyses of seismic waves assuming that they propagated through an earth made up of purely isotropic, linearly elastic material (Section 2.3.9). In such material, the stresses are linearly proportional to the strains via Hooke's law

$$\sigma_{ij} = c_{ijkl} \epsilon_{kl}, \quad (1)$$

and the 81-term tensor of elastic moduli, c_{ijkl} , reduces to two independent elastic constants, λ and μ . As a result, the material's elastic properties are the same in all directions. Although isotropy is a good first approximation in the earth, it is sometimes important to consider deviations from isotropy, or *anisotropy*. In such cases, Hooke's law applies, but the relation between the stresses and strains involves more than two elastic constants. Although there can be up to 21 independent elastic constants, any material in which more than two are needed is called *anisotropic*.

Having more than two elastic constants means that the material's properties differ depending on the direction. Because seismic wave velocities depend on the elastic constants, waves traveling through anisotropic material travel faster or slower depending on their direction, and complicated wave phenomena can occur. For example, a shear wave can be split into two pulses, each with a different polarity and traveling at a different speed (Figs 3.6-1, 2.4-8).

Anisotropy can result from a material's being non-uniform, a condition called *heterogeneity* or *inhomogeneity*. A common situation is when material has directionality in its structure. For instance, plywood is a superposition of thin layers of wood, so its strength (shear modulus) differs in different directions.

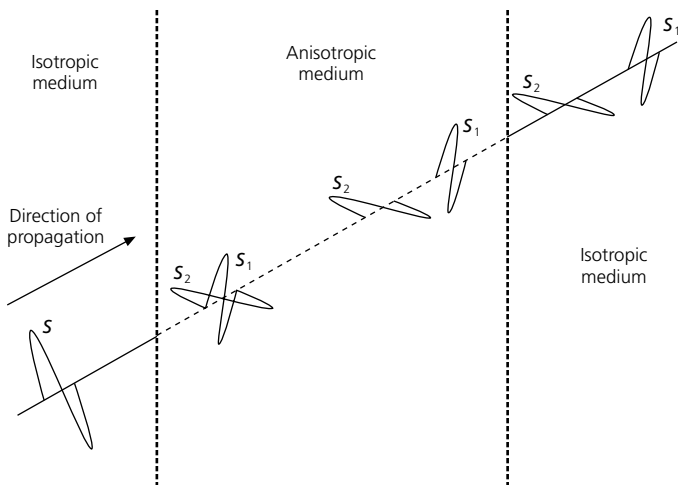


Fig. 3.6-1 Schematic of an initially polarized shear wave split along the fast and slow anisotropic directions, yielding pulses separated in time. The pulses remain split after leaving the anisotropic region.

Similarly, a stack of rock layers with different isotropic velocities can as a whole behave anisotropically, so seismic waves travel with different speeds parallel or perpendicular to the layers. This situation is called shape-preferred orientation (SPO) anisotropy. Anisotropy can also occur for homogeneous materials. For example, the crystal structure of the mineral olivine is homogeneous in that it is composed of the same repeating groups of atoms, but acts anisotropically because its acoustic properties vary in different directions relative to the crystal lattices of an agglomeration of mineral grains. This situation is called lattice-preferred orientation (LPO) anisotropy.

The anisotropic variations of the seismic velocity of earth materials are small compared to the large changes in seismic velocity that occur radially from the surface to the core. Hence, in developing radial models of seismic velocity, anisotropy has traditionally been treated as a secondary effect. Nevertheless, recent efforts to better quantify three-dimensional velocity variations sometimes find that anisotropic perturbations are comparable to lateral velocity changes. It is often difficult, however, to distinguish between the effects of anisotropy and those of heterogeneity. For example, curvature on a refracting interface can simulate many of the effects associated with anisotropy.

An important reason to study anisotropy is that material flow at depth appears to preferentially orient olivine crystals within upper mantle rocks. Hence mapping the seismically "fast" direction lets us investigate the relation between plate motions and mantle flow at depth. Although anisotropy studies are ongoing, and both results and interpretations will change over time, they represent a major frontier in deep earth studies.

3.6.2 Transverse isotropy and azimuthal anisotropy

As discussed in Section 2.3.9, the symmetry of the stress and strain tensors and the idea of strain energy means that no more than 21 of the 81 elastic constants c_{ijkl} are independent. We can thus write the c_{ijkl} tensor as a matrix C_{mn} , where the indices m and n vary from 1 to 6 as the pairs of indices (i, j) or (k, l) take values of (1, 1), (2, 2), (3, 3), (2, 3), (1, 3) and (1, 2), respectively:

$$C_{mn} = \begin{pmatrix} c_{1111} & c_{1122} & c_{1133} & c_{1123} & c_{1113} & c_{1112} \\ c_{2211} & c_{2222} & c_{2233} & c_{2223} & c_{2213} & c_{2212} \\ c_{3311} & c_{3322} & c_{3333} & c_{3323} & c_{3313} & c_{3312} \\ c_{2311} & c_{2322} & c_{2333} & c_{2323} & c_{2313} & c_{2312} \\ c_{1311} & c_{1322} & c_{1333} & c_{1323} & c_{1313} & c_{1312} \\ c_{1211} & c_{1222} & c_{1233} & c_{1223} & c_{1213} & c_{1212} \end{pmatrix} = \begin{pmatrix} C_{11} & C_{12} & C_{13} & C_{14} & C_{15} & C_{16} \\ C_{21} & C_{22} & C_{23} & C_{24} & C_{25} & C_{26} \\ C_{31} & C_{32} & C_{33} & C_{34} & C_{35} & C_{36} \\ C_{41} & C_{42} & C_{43} & C_{44} & C_{45} & C_{46} \\ C_{51} & C_{52} & C_{53} & C_{54} & C_{55} & C_{56} \\ C_{61} & C_{62} & C_{63} & C_{64} & C_{65} & C_{66} \end{pmatrix}. \quad (2)$$